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ORIGINAL ARTICLE / ARTICLE ORIGINAL

In vitro activity of terbinafine against Indian clinical isolates of *Candida albicans* and non-*albicans* using a macrodilution method

Évaluation de l'activité antifongique in vitro de la terbinafine sur des souches cliniques indiennes de *Candida albicans* et non-*albicans* par une méthode de macrodilution

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KEYWORDS
Terbinafine;
Candida;
Antifungal susceptibility;
Macrodilution method

Abstract
Objective. – Terbinafine, an allylamine derivative, is the most representative of this new chemical class of antimycotic compounds. The aim of this study was to determine the anti *Candida* activity of terbinafine compared with amphotericin, fluconazole, ketoconazole and itraconazole.
Method. – The drugs were tested against 25 fresh clinical isolates of *Candida* recovered from blood, urine, sputum and oral and vaginal wash of patients with underlying diseases like cancer, tuberculosis, diabetes, post-operative patients (immunocompromised) and oropharyngeal candidiasis (OPC) or vulvovaginal candidiasis (VVC). The macrodilution method was performed according to the recommendation of National Committee for Clinical Laboratory Standards (NCCLS) M 27-A using RPMI 1640 medium in MOPS buffered to pH 7.1: twofold drug dilution scheme was adapted with the concentration range of 0.03125-64 µg/ml. Yeast inoculum was adjusted to 0.5 McFarland standard with working suspension at 1:100 dilution. The end point for terbinafine was defined as 80% inhibition compared with growth in control wells.
Result. – Under in vitro conditions, terbinafine proved to be highly active against *Candida albicans*, *C. parapsilosis*, *C. guilliermondii*, *C. glabrata* and *C. tropicalis* (minimum inhibitory concentration, MIC range, 0.03125–4 µg/ml) and showed potentially useful activity against

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C. krusei (MIC range, 1–4 µg/ml). Statistically significant difference was observed for terbinafine between *C. parapsilosis*, *C. glabrata* and *C. guilliermondii* and *C. albicans*.
Conclusion. – To the knowledge of the authors, the in vitro data presented in this study is the first from India on terbinafine activity against clinical strains of *C. albicans* and non-*albicans* species with such lower MICs (MIC range 0.03125–4 µg/ml). Thus, in addition to its potent activity against dermatophytes, terbinafine is also very effective against *Candida* species. It could be interesting for clinical laboratories and physician for making appropriate drug choices and patient management decision. In vivo investigations of its antifungal activity against *Candida* should consequently be pursued.

MOTS CLÉS
Terbinafine ;
Candida ;
Sensibilité aux antifongiques ;
Macrodilution

Résumé
Objectif. – Le but de cette étude est d'évaluer l'activité antifongique de la terbinafine, un dérivé des allylamines, sur diverses souches de *Candida* comparativement à l'amphotéricine B, au fluconazole, au kétoconazole et à l'itaconazole.
Méthodes. – Les antifongiques sont testés sur 25 souches cliniques de *Candida albicans* fraîchement isolées du sang, des urines, des expectorations et des lavages buccaux et vaginaux de patients atteints de cancer, de tuberculose, de diabète, de patients postopératoires (immunocompromis) et de candidoses oropharyngées (OPC) ou vulvovaginales (VVC). La méthode par macrodilution en milieu liquide mise en oeuvre selon les recommandations du NCCLS (M 27-A) utilise le milieu RPMI 1640 tamponnée par du MOPS à pH 7.1 ; les gammes de dilutions vont de 64 à 0,03125 µg/ml ; l'inoculum est ajusté à 0,5 unités Mac Farland avec une suspension de travail diluée au 1/100. La dilution limite est définie à 80 % d'inhibition de la culture comparativement au témoin.
Résultats. – Dans les conditions opératoires, la terbinafine est active sur *C. albicans*, *C. parapsilosis*, *C. guilliermondii*, *C. glabrata* et *C. tropicalis* (0,03125 > CMI 4 µg/ml) ; elle est particulièrement active sur *C. krusei* (1 > CMI 4 µg/ml). Des différences significatives sont observées entre *C. parapsilosis*, *C. guilliermondii*, *C. glabrata* et *C. albicans*.
Conclusion. – Selon les auteurs, ce travail sur l'activité in vitro de la terbinafine sur des souches de *C. albicans* et non-*albicans* présentant des CMI aussi basses (0,03125 > CMI 4 µg/ml) constitue la première étude réalisée en Inde. En plus de son pouvoir antifongique reconnu sur les dermatophytes, la terbinafine est aussi très active sur les *Candida*. Cette propriété peut être mise à profit par les laboratoires cliniques et les praticiens pour faire un choix pertinent de l'antifongique à utiliser et pour la prise en charge du patient. L'étude de son activité in vivo dans les candidoses mériterait d'être poursuivie.

Introduction

Candida species are ubiquitous human pathogens, causing localised, invasive or disseminated disease in normal or immunocompromised hosts, promoted by the use of broad spectrum antibiotic, steroids or other immunosuppresive agents, diabetes mellitus, AIDS, cancer chemotherapy and organ transplantation etc. Candidiasis is an infection involving every part of the body and is now the fourth most prevalent organism found in blood stream infections. *Candida albicans* represents more than 80% of isolates from clinical infection [4,25]. Among non-*albicans* species *C. tropicalis* was the commonest isolate (56.2%) followed by *C. guilliermondii* (20.3%), *C. krusei* (13.1%), *C. parapsilosis* (7.4%) and *C. glabrata* (3%) [5,22,33]. These pathogenic species of *Candida* derive their importance not only from the severity of their infections but also from their ability to develop resistance against antifungal drugs. Widespread and prolonged use of azoles has led to the rapid development of the phenomenon of multidrug resistance (MDR), which possess major hurdle in antifungal therapy. The development of allylamine antifungals of which terbinafine is the most effective to date may help to resolve this situation.

Terbinafine, an allylamine derivative is highly active against dermatophytes (*Trichophyton*, *Epidermophyton* and *Microsporum* species) in vitro with minimum inhibitory concentrations (MICs) ranging from 0.001 to 0.01 µg/ml. The compound shows good activity against *Aspergillus* species (MIC range 0.05–1.56 µg/ml), and *Sporothrix schenckii* (MIC range 0.1–0.4 µg/ml), but its ability to inhibit yeasts is still controversial. Earlier susceptibility testing of terbinafine against yeasts showed poor activity, [12,29] while more recent studies have shown that this drug is clearly effective at inhibiting yeasts. [30,32]. This is not surprising since wide differences existed in the methodologies used to evaluate in vitro activity of terbinafine by earlier workers [7,12,27,29]. In contrast to earlier studies, the recently developed National Committee for Clinical Laboratory Standards (NCCLS) M 27-A macrodilution methodology [17] showed a positive activity for terbinafine against yeasts. In this study, we aim to confirm and extend these findings by investigating the in vitro susceptibility of terbinafine against Indian isolates of *Candida* species using the most recently developed macrodilution assay guidelines.

Materials and methods

Study design

The study design followed the recommendations of the document M 27-A (NCCLS) for macrodilution method for determining the in vitro susceptibility pattern of *Candida* spp. against terbinafine compared to amphotericin B and azoles [17].

Test organism

A panel of 25 well characterised pathogenic yeast isolate from the Medical Mycology Laboratory, Department of Biosciences, Jabalpur (M.P.), India, was used in the study. This panel consisted of five strains of *C. albicans*, five strains of *C. parapsilosis*, five strains of *C. tropicalis*, four strains of *C. glabrata*, three strains of *C. guilliermondii*, two strains of *C. krusei* and one strain of *C. vishwanathi*. All strains were primarily recovered from blood, urine, sputum, oral wash and vaginal wash of 25 patients with underlying diseases like cancer, tuberculosis, diabetes, post-operative patients (immunocompromised), oropharyngeal candidiasis (OPC) and vulvovaginal candidiasis (VVC). Before testing, all isolates were subcultured on to SDA slants to ensure optimal growth characteristics.

Reference strains

Quality control and reference strains used in every testing batch are *C. albicans* ATCC 90028, *C. tropicalis* ATCC 750 and *C. glabrata* ATCC 90030. MICs for these reference strains were compared with published control limits and used to guide antifungal susceptibility testing and validation as per NCCLS guidelines [20,23].

Antifungal drugs

The following five antifungal drugs were used. Amphotericin B (AMFOCAN, Dabur India Ltd.), ketoconazole (NIZRAL; Johnson and Johnson) Janssen Pharmaceutica, Registered Trademark of Johnson and Johnson, USA), itraconazole capsules (CANDISTAT; E Merk India Ltd.; Licensed user of T.M.), Fluconazole (Flustan™; Dr. Reddy's Lab. Ltd.)™ Trademark under registration and terbinafine (DASKIL); Novartis India Ltd.).

Drug dilutions

In this study drug dilutions were prepared according to NCCLS M 27-A protocol [17]. Stock solution of drug (100×) were prepared in 100% dimethyl sulphoxide (DMSO) (Sigma Chemicals Co., St. Louis, MO) for (itraconozole, ketoconazole and amphotericin B) or sterile distilled water (fluconazole) or Polyethylene glycol (PEG)-400 (Union Carbide, Danbury, Conn) for terbinafine immediately prior to use. From these stock solutions intermediate test drug dilutions were prepared which is to be 10× the strength of the final drug concentrations, with RPMI 1640 medium as diluent (by two-fold drug dilution scheme); NCCLS standards (Approved M 27-A) of macrodilution broth reference method across the concentration range (0.3125–640 μg/ml).

Inoculum preparation

All isolates were subcultured at least twice to ensure purity and viability. Yeasts were grown on Sabourauds Dextrose Agar (SDA) plates for 24 h at 35 °C. Five colonies each at least 1 mm in diameter were suspended in 5 ml of sterile 0.85% saline. The resulting yeast suspension was mixed for 15 s with a vortex. The turbidity of the suspension was adjusted spectrophotometerically to match the transmittance of a 0.5 McFarland barium sulphate turbidity standard at 530 nm. This procedure yielded a yeast stock suspension of 1×10^6 to 5×10^6 cells per ml [21]. A working suspension was made by 1:100 dilution followed by 1:20 dilution of the stock suspension with RPMI 1640 broth medium, which results in 0.5×10^3 to 2.5×10^3 cells per ml. Test inocula were made in sufficient volumes to directly inoculate each MIC tube with 0.9 ml.

Test medium

Media used was RPMI 1640 (10.3 g/l) with L-glutamine without bicarbonate (Himedia, India) buffered to pH 7.0 with MOPs buffer (3-*N*-morpholino propane sulphonic acid) to a final molarily 0.165 M.

Susceptibility testing procedure

The 10× drug dilution were dispensed in 0.1 ml volumes into sterile glass bottles. Each bottle was then inoculated by 0.9 ml volumes of diluted yeast inoculum suspension. This step brought the drug dilutions to the final test drug concentration (64–0.03125 μg/ml) for all five antifungal drugs. The growth control tube (s) contained a 0.9 ml volume of an inoculum supension and a 0.1 ml volume of drug free medium. Quality control organism was tested in the same manner as the other isolates and was included each time with the tested batch. In addition, 1 ml of uninoculated drug free medium was included as a sterility control. All bottles were incubated at 35 °C and MICs were read after 24 and 48 h of incubation according to the NCCLS M 27-A protocol [19].

Interpretation of results

MIC is defined as the lowest concentration in micrograms/millilitre (μg/ml) of an antifungal that substantially inhibits 80% (azoles) and 100% (amphotericin B) growth of the organism in comparison with drug free control growth. For terbinafine, 80% inhibition of growth has been accounted as the assay end point in this study. The tested strains were categorised into three groups: Susceptible (S), Susceptible Dose Dependent (SDD) and Resistant (R) depending on the MIC of different antifungal agents according to the following interpretative break points: for amphotericin B ≤ 0.25 µg/ml (S), 0.5 µg/ml (SDD), ≥ 1 µg/ml (R), for fluconazole ≤ 8 µg/ml (S), 16–32 µg/ml (SDD), ≥ 64 µg/ml (R), for itraconazole ≤ 0.125 µg/ml (S), 0.25–0.5 µg/ml (SDD), ≥ 1 µg/ml (R), for ketoconazole ≤ 0.0625 µg/ml (S), > 0.125 µg/ml (R) [18,24,25]. Clinically relevant break points are currently not available for terbinafine but a correlation between MIC and out come of treatment provide interpretative breakpoint similar to fluconazole [32,36]. Sensitive, Susceptible Dose Dependent and Resistant categorisation were done on the basis of the drug concentration that can easily be achieved in the blood following administration, the drug concentration that would be achieved with higher doses and the concentration that cannot be achieved or of high toxicity, respectively.

Statistical analysis

Student “*t*” test was employed to determine the significance of the difference between the geometrical means of MICs values. Statistically significance was set at $P < 0.05$.

Result

The results of the present study revealed that the antifungal terbinafine was found to be highly active against all *Candida* species (Table 1). Geometric mean MIC concentrations were observed as 0.35 (+ 1.38) for *C. parapsilosis*, *C. glabrata*, *C. guilliermondii* followed by 1.41 (± 0.70) for *C. tropicalis*, 2 (± 1.52) for *C. krusei* 2.83 (± 1.41) for *C. albicans*. Statistically significant difference was observed for terbinafine in *C. parapsilosis*, *C. glabrata*, and *C. guilliermondii* and these were significantly lower compared with *C. albicans* (2.83 mg/l verus 0.35), other findings showed insignificant differences. In this study, this drug was found to be in active only against *C. vishwanathi* but showed potentially interesting activity against intrinsically resistant *C. krusei* (MIC of 1–4 µg/ml. All 25 *Candida* spp. 25 (100%) were found to be susceptible (S) (MIC range 0.03125–0.25 µg/ml) to amphotericin B, and 19 isolates (76%) showed closer competition with terbinafine (MIC range 0.03125–4 µg/ml) in this assay but azoles showed MICs over the full test range for fluconazole (0.25–64 µg/ml), ketoconazole (0.03125–64 µg/ml) and itraconazole (0.3125–8 µg/ml). Among azoles, 14 (56%), 10 (40%) and 9 (36%) isolates were found to be susceptible to itraconazole, ketoconazole and fluconazole, respectively (Table 2 and Fig. 1) for each azole. There was a small sub population of isolates which displayed higher MICs and showed SDD and resistant category while terbinafine gave susceptible category to the same isolates with lower MICs in this assay. Isolates of *Candida* showing reduced susceptibility to azole were found to be susceptible to terbinafine. Comparative susceptibility pattern of five antifungal agents showed that terbinafine took 2nd and 3rd position in the order of susceptibility and gave a very close competition with polyenes and azoles. MICs values obtained by the reference strains with antifungal agents were in agreement with the NCCLS recommended ranges (Table 3).

Discussion

In the present study terbinafine showed its remarkably brilliant antifungal activity against *C. albicans* and all non-*albicans* species (MIC range, 0.03125–4 µg/ml) isolated from oral wash, vaginal wash , urine and blood sample of OPC, VVC (Superficial candidiasis), immunocompromised and cancer patients (Systemic mycoses). Our data show some variance with those of earlier investigators [29,32] who were generally unable to show significant in vitro activity for terbinafine against all *Candida* species, except for *C. parapsilosis*. Interestingly, in the present study the MICs of terbinafine against *C. albicans* and non-*albicans* were much lower than those obtained by earlier workers [7,9,11,19,31,34]. They reported MICs of around (25 µg/ml) for *C. albicans* and MIC range (10.0 – > 128 µg/ml) for non-*albicans* spp. while in the present study we are reporting MIC range (2–4 µg/ml) and MIC range (0.03125–4 µg/ml) for *C. albicans* and non-*albicans*, respectively.

This may be likely due to two characteristics of NCCLS method, which provide good results for terbinafine. Firstly, the readout is 80% growth inhibition, eliminating the uncertainty of trailing end points seen in *Candida* treated with dilution series of azoles or allylamines. Secondly, the NCCLS medium is buffered at neutral pH, while earlier methods used unbuffered media which are rapidly acidified by *Candida* species [11,31,33]. Terbinafine is much less active at low pH [29] so that inclusion of a neutral buffer is an essential prerequisite for testing this and related drugs against yeasts. Ryder and Dupont [28] also showed that terbinafine has optimal antifungal activity when used in an assay with a neutral pH range and diminishes under acidic

Table 1 Terbinafine MICs from this study against 25 isolates of *Candida* spp. in comparison with previously published values
Tableau 1 CMI de la terbinafine évaluée sur 25 souches de *Candida* sp. en comparaison avec les CMI observées dans la littérature

Organism	Number of isolates tested	This study MIC	Other studies MIC	References
C. albicans	5	2–4	0.03- > 128.0	[30,7,12,34,9,29]
C. parapsilosis	5	0.03125–4	1.2- > 10.0	[30,7,12,34,9,29]
C. tropicalis	5	1–2	1.2 - > 128.0	[30,7,12,34,9,29]
C. glabrata	4	0.03125–4	10.0- > 128.0	[30,7,12,34,9,29]
C. guilliermondii	3	0.03125–4	0.8- > 128.0	[30,7,12,34,9,29]
C. krusei	2	1–4	10.0 – 100.0	[30,7,12,34,9,29]
C. vishwanathi	1	> 64	NR	

NR: not reported.

Table 2 In vitro susceptibility of clinical isolates of *Candida* spp. against five antifungal drugs determined by a macrodilution method [a] (48 h)
Tableau 2 Sensibilité in vitro de souches cliniques de *Candida* sp. à 5 antifongiques par une méthode de macrodilution [a] (48 h)

Antifungal Drug	*C. albicans* *(5)*	*C. parapsilosis* *(5)*	*C. tropicalis* *(5)*	*C. glabrata* *(4)*	*C. gulliermondii* *(3)*	*C. krusei* *(2)*	*C. vishwanathi* *(1)*
Amphotericin B							
Range [b]	(0.0625-0.25)	(0.0625-0.025)	(0.03125-0.125)	(0.0625-0.25)	(0.03125-0.0625)	(0.25)	0.0625
Geometric mean [c]	0.125	0.125	0.0625	0.125	0.0441	(0.25)	0.0625
MIC 80 [d]	0.125	0.25	0.125	0.125	-	-	-
Category [e]	S	S	S	S	S	S	S
Mode (0.0625) [f]							
Fluconazole							
Range [b]	(0.25-16)	(4-16)	16	(16-64)	(0.5-64)	64	0.25
Geometric mean [c]	2	8	16	32	5.65	64	0.25
MIC 80 [d]	1	16	16	64	-	-	-
Category [e]	S/SDD	S/SDD	SDD	SDD/R	S/SDD/R	R	S
Mode (16) [f]							
Ketaconazole							
Range [b]	(0.03125-16)	(0.03125-8)	(0.03125-64)	(0.03125-8)	(0.125-8)	0.125	0.125
Geometric mean [c]	0.707	0.5	1.41	0.5	1	0.125	0.125
MIC 80 [d]	8	8	0.0625	8	-	-	-
Category [e]	S/SDD/R	S/SDD/R	S/SDD/R	S/SDD/R	S/R	S	S
Mode (0.03125) [f]							
Itraconazole							
Range [b]	(1-8)	(0.03125-0.0325)	(0.03125-1)	(0.0625-4)	(0.03125-4)	(0.0625-2)	0.125
Geometric mean [c]	2.82	0.0441	0.176	0.5	0.353	0.353	0.125
MIC 80 [d]		2	0.0625	0.0625	2	-	-
Category [e]	R	S	S/R	S/R	S/R	S/R	S
Mode (0.03125) [f]							
Terbinafine							
Range [b]	(2-4)	(0.03125-4)	(1-2)	(0.03125-4)	(0.03125-4)	(1-4)	-
Geometric mean [c]	2.82	0.353	1.41	0.353	0.353	2	2.82
MIC 80 [d]	4	4	2	4	-	-	-
Category [e]	S	S	S	S	S	S	R
Mode (4) [f]							

[a] NCCLS, National Committee for Clinical Laboratory Standards (M27-A).
[b] Range, range of MICs.
[c] GM, geometric mean.
[d] MIC 80, MIC at which 80% of the isolates are inhibited.
[e] Category; tested strains were categorised into three groups: resistant (R), sensitive (S) and Susceptible Dose Dependent (SDD).
[f] Mode, modal MIC of 25 *Candida* spp. for individual drug.

Table 3 Recommended MIC limits for reference strains against antifungal agents for broth macrodilution method (NCCLS: M 27-A)
Tableau 3 Valeurs limites des CMI des antifongiques pour les souches de référence pour la méthode de macrodilution (NCCLS: M 27-A)

Organism	Amp-B	Flu	Keto	Itra	Ter *
C. albicans MTCC 3017/ATCC 90028	0.5-2.0	0.25-1.0	0.06	0.125	1
C. tropicalis MTCC 184/ATCC 750	0.5-2.0	1.0-4.0	0.06	0.125	> 128
C. glabrata MTCC 3019/ATCC 90030	0.25-1.0	8-32			

ATCC (R) is a registered trademark of the American type culture collection. MTCC is a registered trademark of Microbial type culture collection. • No established MIC limits yet published for terbinfine but different data presents MIC limits by applying NCCLS M27-A dilution assay to terbinafine for reference strains.

condition. In the present study terbinafine showed its fungisitic activity against *C. albicans* and fungistatic activity against *C. parapsilosis* similar to the results obtained by earlier workers [6,11].

The in vitro activity of terbinafine against *Candida* species is thus of interest with regard to potential clinical efficacy. Topical applied terbinafine was found to be highly effective in cutaneous candidiasis [32]. Oral terbinafine has also been reported to show efficacy in cutaneous candidiasis. After oral administration (250 mg/day) terbinafine attains peak levels of up to 12 μg/ml in the stratum corneum [6]. Pharmacokinetic data for humans indicate that after a single dose (Oral dose) of 500 mg of terbinafine, peak drug concentration in plasma of 2 μg/ml are reached

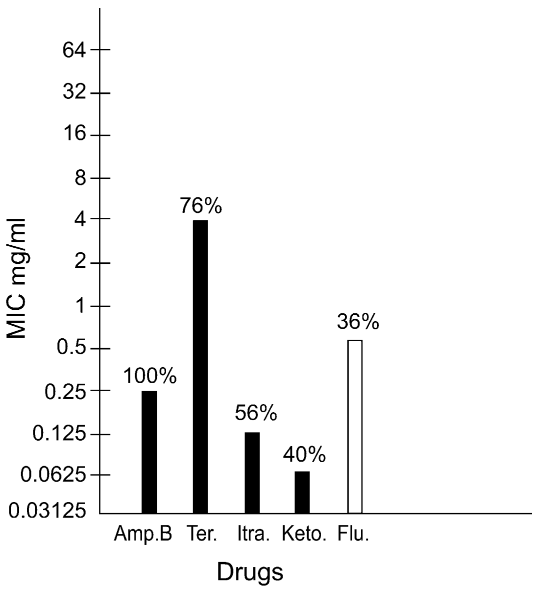

Figure 1 Showing comparative susceptibility pattern of different *Candida* spp. against five antifungal agents with respect to MICs.

Figure 1 Sensibilité comparée de différentes espèces de *Candida* à cinq antifongiques.

within 2 h and the elimination half life is 11.3 h [36,37]. Comparatively, in the present study our data showed MICs below 2 μg/ml (MIC range 0.03125–4 μg/ml), which suggest that terbinafine could be in a promising new agent for treating case of candidiasis.

Our comparative in vitro data of terbinafine with amphotericin B and azoles indicate that terbinafine is in close competition with amphotericin B and produce much lower MIC value for some isolates of *Candida* showing decreased susceptibility to azoles in view of its, various investigators have reported successful antifungal treatment of superficial and systemic candidiasis by using terbinafine in combination with amphotericin B and azoles [1–3,8,13,15,16,26,27, 35] in vitro activity of terbinafine in combination with azoles and polyens could be judged to prove their synergetic effect. Thus, the therapeutic potential of terbinafine extends well beyond its current use in acute and chronic dematophytes to cure wide range of cutaneous, subcutaneous and systemic mycoses due to *Candida* sp. alone or in combination with azoles against clinical isolates of *Candida* sp. with decreased susceptibility to azoles [1,1014,15]. The in vitro susceptibility of *Candida* species to terbinafine has already been described by different investigators [12,30]. However, there is no information available from India, in this regard as ecology of *Candida* species and the epidemiology of candidiasis are different in many parts of the world, it is important to know whether such differences also reflect variations in susceptibilities to terbinafine [29, 30].

The availability of reproducible antifungal susceptibility testing methods, now permits analysis of data in vitro and with outcome in vivo, vital for meaningful communication between clinical laboratories and physician for making appropriate drug choices and patient management decision. The in vitro data presented for terbinafine in this study was considered to be the first published data from India against clinical isolates of *Candida* spp. Therefore investigations concerning its antifungal activities in vivo against such organism should be pursued.

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References

[1] Barchiesi F, Scalise G. In vitro activities of terbinafine in combination with fluconazole and itraconazole against *Candida albicans* with reduced susceptibility to azoles. Antimicrob Agents Chemother 1997;41:1812–4.

[2] Canton E, Peman J, Gobernado M, Viudes A, Espinel-Ingroff A. Synergy activities of fluconazole and voriconazole with terbinafine against four *Candida* species determined by Checkboard, Time-kill and Etest Methods. Antimicrob Agents Chemother 2005;49:1593–6.

[3] Cuenca-Estrella M. Combination of antifungal agents in therapy-what value are they? J Antimicrob Agents Chemother 2004;54:854–69.

[4] Edmond MB, Wallace SE, Mc Clish DK. Nosocomial blood stream infection in United States hospitals, 3-year analysis. Clin Infect Dis 1999;29:239–44.

[5] Eiteon Abu HK, Eikarmi AZ. Isolation and characterization of *Candida* species in Jordian cancer patients; prevalence, pathogenic determinants and antifungal sensitivity. Jpn J Infect Dis 2004;57:279–84.

[6] Faergemann J, Zehender H, Denoucl J. Level of terbinafine in plasma, stratum Corneum, Dermis, Epidermis, Sebum, hairs and nails during and after 200 mg terbinafine one/day for 4 weeks. Acta Derm Venereol 1993;73:305–9.

[7] Gabor P, Josef G, Hubet M. Antifungal activity of the allylamine derivative terbinafine in vitro. Antimicrob Agents Chemother 1987;31:1365–8.

[8] Ghannoum MA, Elewski B. Successful treatment of fluconazole resistant OPC by a combination of fluconazole and terbinafine. Clin Diagn Lab Immumol 1999;6:921–3.

[9] Govdard MY, Buffard H, Ferrai PR. In vitro spectrum of action of new antifungal derivative of notified; terbinafine (SF 86-327). Pathol Biol 1986;34:680–3.

[10] Hay RJ. Therapeutic potential of terbinafine in subeutaneous and systemic mycosis. Br J Dermatol 1999;41:36.

[11] Hazen KC. Fungicidal versus fungistatic activity of terbinafine and itraconazole; an in vitro comparison. J Am Acad Dermatol 1998;38:37–41.

[12] Hiratani T, Asagi Y, Yamguchi H. Evaluation of in vitro antimycotic activity of terbinafine, a new allylamine agent. Jpn J Med Mycol 1991;32:323–32.

[13] Johnson MD, MacDougall C, Ostrosky-Zeichner L, Perfect JR, Rex JH. Combination antifungal therapy. Antimicrob Agents Chemother 2004;48:693–715.

[14] Luna B, Drew RH, Perect JR. Agents for treatment of invasive fungal infections. Otolaryng Clin N Am 2000;33:277–99.

[15] Cuenca-Estrella M. Combination of antifungal agents in therapy what value are they? J Antimicrob Agents Chemother 2004;54:854–69.
[16] Mukherjee PK, Sheehan DJ, Hitchcock CA, Ghannoum MA. Combination treatment of invasive fungal infections. Clin Microbiol Rev 2005;18:163–94.
[17] National Committee for clinical Laboratory Standard (NCCLS). Refrence method for Broth dilution antifungal susceptibility testing of yeasts (M 27-A). Wayne, PA: Approved Standards; 1997.
[18] Nguyen MH, Clanes CJ. Do in vitro susceptibility data predict the microbiologic response to Amp. B? Results of a prospective study of patients with *Candida* fungirmia. J Infect Dis 1998; 177:425–30.
[19] Perez A. Terbinafine: a blood new spectrums of indication in several subcutaneous and systemic and parasitic disease. Mycoses 1999;42:111–4.
[20] Pfaller MA, Espinel Ingroff A, Rex JH. Quality control guidelines for NCCLS recommended broth macrodilution testing of Amp. B, fluconazole and flucytosine. J Clin Microbiol 1995: 1104–7.
[21] Pfaller MA, Rinaldi MC, Bartlett MS. Multicenter evaluation of 4 methods of yeast inoculum preparation. J Clin Microbiol 1988;26:1437–41.
[22] Pfaller MA, Diekewa DJ. The role of sentinel surveillance of candidemia: trends in species distribution and antifungal susceptibility. J Clin Microbiol 2002;40:3551–7.
[23] Rex JH, Pfaller MA. Quality control guidelines for NCCLS recommended broth macrodilution testing of ketoconazole and itraconazole. J Clin Microbiol 1996:816–7.
[24] Rex JH, Pfaller MA, Galgiani JN, NCCLS. Development of interpretative break points for antifungal susceptibility testing conceptual framework and analysis of in vitro–in vivo correlation data for fluconazole, itraconazole and *Candida* infection. Clin Infect Dis 1997;24:235–47.
[25] Rodriquez-Tudela JL, Martinez-suarex JV. Correlation of in vitro susceptibility test results with in vivo response; a study of azole therapy in AIDS patients J. Antimicrob Agents Chemother 1995;35:793–804.
[26] Rothe A, Seibolel M, Cornely OA. Combination therapy of disseminated *Fusarium oxysporum* infection with terbinafine and amphotericin B. Ann Hematol 2004;83:394–7.
[27] Rubio MC, Gil J, Rezusta A, Bueno MR. In vitro evaluation of 3 new antifungal agents: amorlfine (RO 14-4767/002), terbinafine (SF 86–327) and itraconazole (R 51–211). Rev Esp Quimioterap 1993;8:125–39.
[28] Ryder NS, Dupont MC. Inhibition of squalene epoxidase by allylamine antimycotic compounds: a comparative study of the fungal and mammalian enzymes. Biochem J 1985;230: 765–70.
[29] Ryder NS. Specific inhibition of fungal sterol biosynthesis by SF 86–327, a new allylamine antifungal agent. Antimicrob Agents Chemother 1985;27:252–6.
[30] Ryder NS, Favre B. Antifungal activity and mechanism of action of terbinafine. Rev Comp Pharmacother 1997;8:275–87.
[31] Ryder NS, Mieth H. Allylamine antifungal drugs. Curr Top Med Mycol 1992;4:158–88.
[32] Ryder NS, Wayner S. In virto activities of terbinafine against cutaneous isolates of *C. albicans* and other pathogenic fungi. Antimicrob Agents Chemother 1998;42:1057–61.
[33] Saurla Z, Yardeni M. Species distribution of antifungal susceptibility of *Candida* blood stream isolates in a tertiary medical center in Israel. Eur J Clin Microbiol Infect Dis 2005;24: 592–5.
[34] Shadomy S, Espinell Ingroff A, Gebhart RJ. In vitro studies with SF 86–327; a new orally active allylamine derivative. Sabouraudia 1985;23:125–32.
[35] Sofia P, Gloria G, Rinald MG. In vitro activities of terbinafine incombination with fluconazole, itraconazole, voriconazole and posaconazole against clinical isolates of *Candida glabrata* with decreased susceptibility to azoles. J Clin Microbiol 2002; 40:1831–3.
[36] Stutz A, Petranyl G. Synthesis and antifungal activity of (*E*)-*N*-(6-6-dimethyl-2-hepten-4-ynyl)-*N*-methyl-1-napthalene-methanamine (SF 86–327) and related allylamine derivatives with enhanced oral activity. J Med Chem 1984;27:1539–43.
[37] Zaias N. *Candida*: a review of clinical experience with Lamisil (R). Dermatology 1997;194(Suppl 1):10–3.